

Week 4 — Reproducible Geospatial Analysis Workflow

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1. Methodology

1.1 Data Sources

Building footprints loaded in Week 3 from two open datasets:

Source	Collection	Documents	License
Microsoft Global Building Footprints	buildings_ms	1,763,356	ODbL
Google Open Buildings v3	buildings_google	2,691,812	CC BY-4.0 / ODbL

Municipality boundaries (169 PDET polygons) in upme.municipalities.

1.2 Aggregation Pipeline

Each building collection is aggregated independently with the following MongoDB aggregation pipeline:

```
$match → {divipola: {$ne: null}}
$group → by divipola: $sum(count), $sum(area_sqm), $avg(area_sqm), $percentile(area_sqm, 0.5)
$lookup → municipalities (for name, department)
$unwind → flatten lookup array
$project → shape to results.schema.json
```

Each result document is upserted into upme.results with unique key (divipola, source), producing one row per municipality per dataset.

1.3 Area Computation

Building areas (area_sqm) were pre-computed at load time (Week 3) using EPSG:9377 (MAGNA-SIRGAS Origen Nacional), Colombia's official equal-area projected CRS. This avoids latitude-dependent distortion inherent in WGS84 geographic coordinates. Geometries are stored in EPSG:4326 for MongoDB's 2dsphere index compatibility.

1.4 Median via MongoDB 7 \$percentile

MongoDB 7.0 introduced \$percentile as a group accumulator. The pipeline uses method: "approximate" (t-digest algorithm) to compute median area without sorting the full dataset in memory. This is bounded-memory and accurate within ~1% of the true median.

1.5 Sensitivity to Google Confidence Threshold

Google Open Buildings assigns a confidence score in [0.6, 1.0] to each detection. The pipeline was re-run at thresholds [0.6, 0.65, 0.7, 0.75, 0.8, 0.9] to measure how building counts and total area change. Microsoft does not provide a confidence score (all detections included).

2. Aggregation Results

2.1 National Totals

Metric	Microsoft	Google	Ratio (G/MS)
Buildings	1,763,356	2,691,812	1.53x
Total rooftop area (km²)	221.18	222.90	1.01x
Mean building area (m²)	114.55	80.61	0.70x
Municipalities with data	169	165	—

Google detects significantly more buildings than Microsoft, but at smaller mean footprint sizes. Total rooftop area converges between sources, indicating both capture similar aggregate coverage despite different detection sensitivities.

2.2 Per-Municipality Results (Top 20)

DIVIPOL	Name	Department	MS count	MS area km²	MS mean	MS median	Google count	Google area km²	Google mean	Google median
47001	SANTA MARTA	MAGDALENA	67,435	14.18	210.21	100.49	183,114	16.00	87.39	58.45
20001	VALLEDUPAR	CESAR	64,088	13.53	211.07	92.68	171,908	15.41	89.65	55.96
76109	BUENAVENTURA	VALLE DEL CAU	34,382	7.27	211.34	93.29	104,324	7.70	73.82	52.18
18001	FLORENCIA	CAQUETÁ	26,757	6.67	249.40	106.64	57,994	7.35	126.81	84.03
05837	TURBO	ANTIOQUIA	42,642	4.59	107.66	72.35	77,890	5.96	76.56	58.40
52835	SAN ANDRES DE TU	NARIÑO	41,294	4.51	109.21	76.94	49,702	3.71	74.65	60.13
19698	SANTANDER DE QU	CAUCA	30,873	4.24	137.50	92.39	35,527	3.85	108.35	75.39
54810	TIBÚ	NORTE DE SAN	35,339	3.55	100.55	68.61	32,767	2.65	80.80	59.65
18753	SAN VICENTE DEL C	CAQUETÁ	22,363	3.54	158.45	95.70	22,232	2.67	119.91	74.96
81065	ARAUQUITA	ARAUCA	29,059	3.46	119.13	80.58	36,164	3.02	83.54	59.06
05045	APARTADÓ	ANTIOQUIA	10,304	3.45	334.41	96.00	37,193	3.70	99.45	66.51
19256	EL TAMBO	CAUCA	37,012	3.29	89.02	68.65	43,200	2.71	62.70	45.88
95001	SAN JOSÉ DEL GUA	GUAVIARE	21,535	3.11	144.36	86.13	32,798	3.17	96.72	65.67
81736	SARAVENA	ARAUCA	17,499	3.08	176.04	86.93	27,625	2.19	79.32	58.56
86568	PUERTO ASÍS	PUTUMAYO	20,266	2.96	146.17	86.29	26,218	2.56	97.81	73.43
23807	TIERRALTA	CÓRDOBA	29,628	2.93	98.73	74.61	42,469	3.04	71.60	55.50
81794	TAME	ARAUCA	20,687	2.73	131.92	82.21	35,728	3.18	89.10	61.71
23466	MONTELÍBANO	CÓRDOBA	21,096	2.70	128.00	79.66	30,083	2.66	88.34	64.71
19142	CALOTO	CAUCA	16,173	2.33	143.95	88.18	24,041	2.79	116.13	71.23
19548	PIENDAMÓ	CAUCA	17,339	2.29	131.93	88.39	31,395	2.71	86.47	55.13

Full results: [data/processed/results_summary.csv](../data/processed/results_summary.csv) (334 rows).

2.3 Cross-Source Divergence

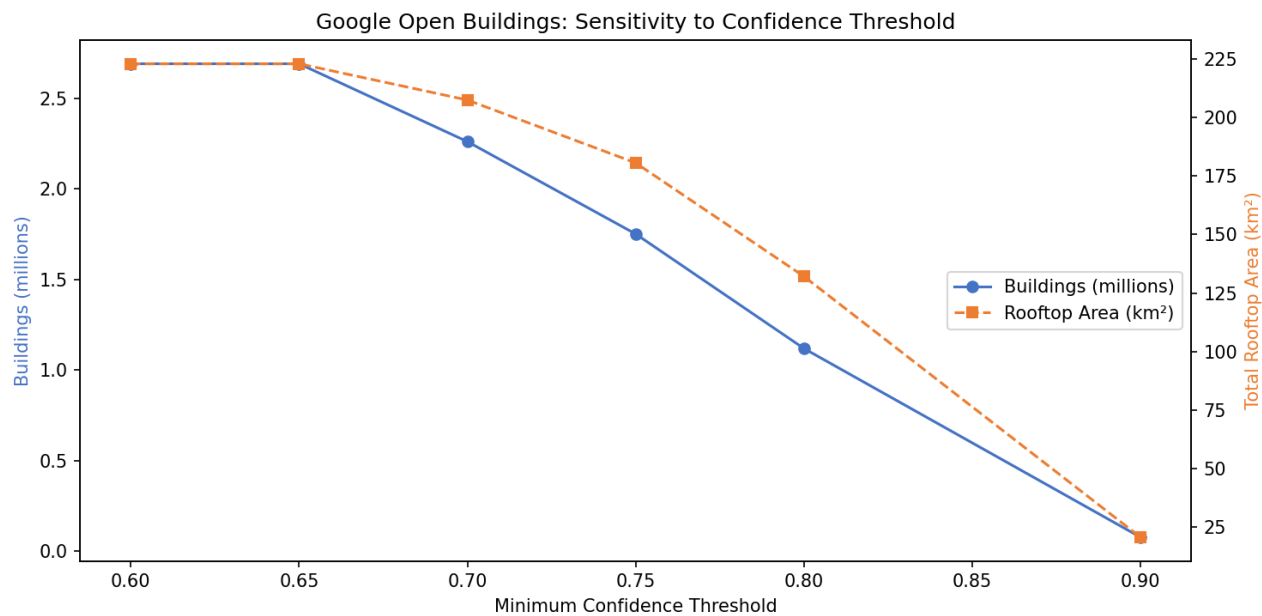
- Municipalities with MS data: 169
- Municipalities with Google data: 165
- Municipalities with both: 165
- Municipalities where counts differ > 20%: 126 (76%)

The high divergence rate confirms the project mandate to compare both datasets rather than relying on a single source.

3. Sensitivity Analysis

Impact of increasing the minimum confidence threshold on Google results:

Min confidence	Munis with data	Buildings	Total area (km²)
0.60	165	2,691,812	222.90
0.65	165	2,691,812	222.90
0.70	165	2,261,910	207.42
0.75	164	1,751,854	180.57
0.80	164	1,119,363	132.10
0.90	163	77,003	20.76



Raising the threshold from 0.6 to 0.7 reduces building count while preserving most of the total rooftop area, suggesting that low-confidence detections tend to be small structures. A threshold of 0.7 offers a reasonable accuracy-coverage tradeoff.

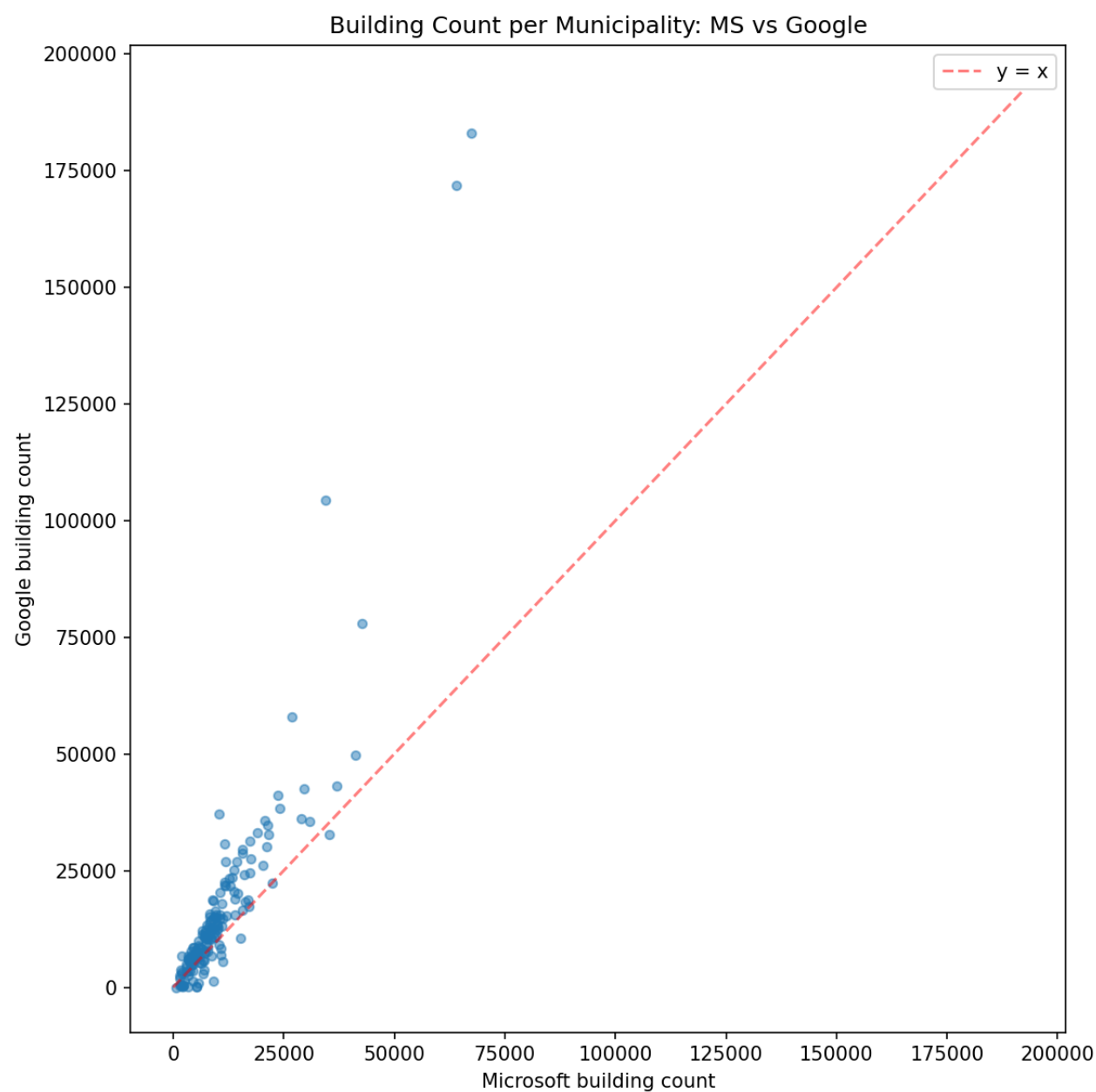
4. Visualizations

4.1 Choropleth Maps

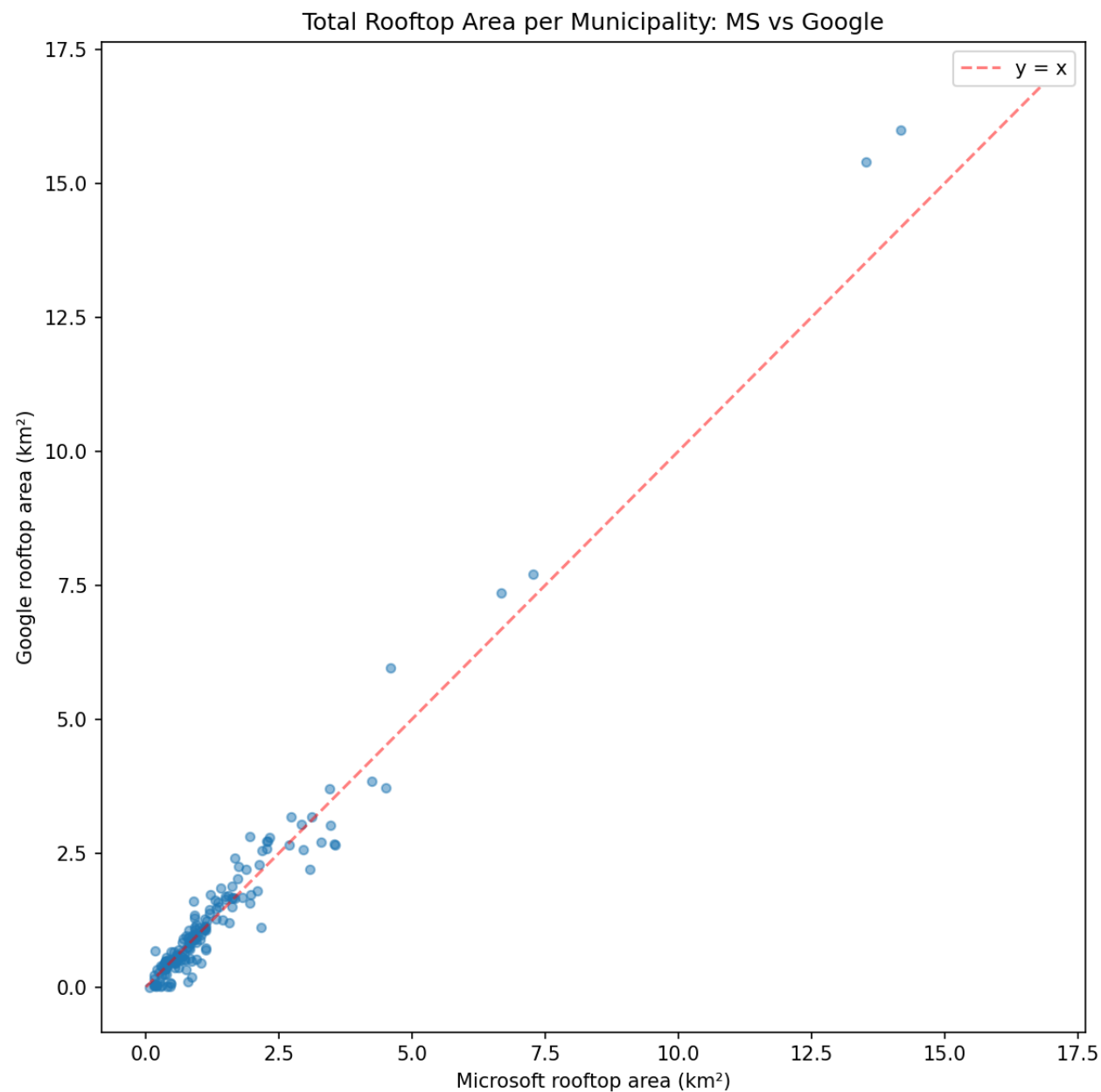
Interactive Folium maps with hover tooltips:

Map	File
MS building count	map_microsoft_building_count.html
MS rooftop area	map_microsoft_rooftop_area.html
Google building count	map_google_building_count.html
Google rooftop area	map_google_rooftop_area.html

4.2 Cross-Source Comparisons

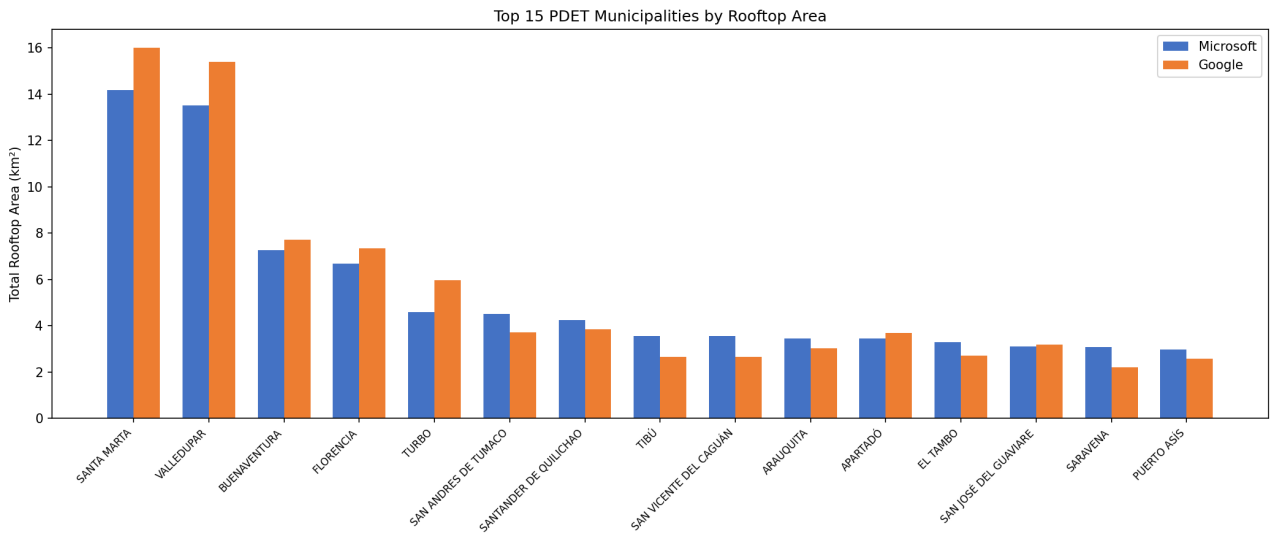


Points above the red $y=x$ line indicate municipalities where Google detects more buildings than Microsoft (majority of cases).



Rooftop area shows tighter agreement between sources than raw building counts, as Google's extra detections tend to be small structures.

4.3 Top 15 Municipalities



5. Reproducibility

```
# Prerequisites: MongoDB running, venv activated, Week 3 data loaded
source .venv/bin/activate

# Run the full analysis workflow
MONGO_URI="mongodb://localhost:27017/" python scripts/analyze.py

# Output files:
# upme.results          — 334 documents (169 MS + 165 Google)
# data/processed/results_summary.csv
# data/processed/sensitivity_google.csv
# docs/week4-report.md
# docs/map_*.html      — 4 choropleth maps
# docs/chart_*.png     — 4 comparison charts
```

6. Known Limitations

- San José de Uré (23580): missing from municipality polygons (GADM 4.1 limitation, documented in Week 3). Buildings in this territory are absent.
- Approximate median: \$percentile uses t-digest; exact median would require sorting the full collection per group.
- Google confidence floor: all detections have confidence ≥ 0.6 . The sensitivity analysis cannot assess detections below this threshold.
- Area source difference: Microsoft areas are computed via EPSG:9377 reprojection; Google areas come from the source CSV area_in_meters field (satellite-derived). Minor methodological differences are expected.